

Article

The Lateral Growth of Branches into Small Canopy Gaps: Implications for Competition between Canopy Trees

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Abstract: Much research in forest ecology has been devoted to examining the effect of gap formation on regeneration dynamics. However, comparatively little research has examined the process of gap closure, in which larger trees bordering the gap grow laterally to exploit available light. Thus, it remains uncertain whether disturbance disrupts or reinforces the competitive hierarchy established among different species and sizes classes. We quantified the lateral growth of three hardwood tree species with differing autecologies both before and after the formation of small gaps created by single-tree selection. Linear mixed-effect models were employed to link lateral growth to species and stem diameter to examine whether gap formation favors intolerant species and small trees in the canopy. Additional models were also developed to examine the relationship of lateral growth with branch length and tree height. Before gap formation, the mid-tolerant yellow birch grew considerably faster than the tolerant sugar maple and American beech. However, yellow birch was less responsive to gap formation (~16%) than sugar maple or beech, whose lateral growth increased by 42% and 39%, respectively. This suggests that gap formation reinforces the competitive dominance of tolerant species. In contrast, gap formation disrupts the competitive dominance of large trees in the canopy, since the lateral growth of small trees increased five times that of large trees. Thus, small silvicultural gaps bordered by small trees may close too quickly to permit the regeneration of mid-tolerant species. Following the release, small trees also grew faster than their larger counterparts, suggesting that lateral growth declines as the cost of reproduction increases with tree size. However, lateral growth did not vary with tree height or branch length, suggesting that lateral growth does not decline due to increasing support costs or hydraulic limitation.

Keywords: disturbance; gap formation; gap closure; competitive interaction; shade tolerance



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1. Introduction

The formation of canopy gaps can exert a profound influence on subsequent stand dynamics by altering competitive interactions among the remaining trees at a neighborhood scale. Much research in forest ecology has been devoted to studying gap dynamics, especially the effect of gap formation on local forest regeneration dynamics. For example, numerous studies have examined how increased plant resource availability in canopy gaps can stimulate the establishment of juvenile plants and the release of advanced regeneration within gaps and their periphery [1–4]. However, comparatively few studies have examined the process of gap closure [5,6], in which larger trees bordering the gap expand their crowns laterally to exploit the increase in light availability, despite the wide recognition that lateral gap closure profoundly influences future stand structure and composition [7–10]. Thus, less is understood about how canopy gap formation influences the growth and competitive interactions among larger edge trees.

Several studies have examined the response of residual canopy trees to gap-scale disturbance by measuring subsequent patterns in stem growth [10–13]. Jones et al. [12]

measured the increase in basal area increment following single-tree selection in tolerant hardwood stands and found that tree size was the most important factor determining the magnitude of the response, with small trees responding more than large trees. Similar diameter growth responses to gap creation have been found in western conifer stands, with the greatest growth differences ranging from 39% for large trees to above 100% for small trees [13]. This suggests that disturbance can disrupt the inherent asymmetry of competition between large and small trees, allowing overtopped trees to exploit the increase in light availability by growing laterally into new gaps.

Comparatively little research has examined branch growth response of gap-edge trees to gap formation from selection silviculture [1,7–9]. Most of the studies that exist have reported that the branches of small trees laterally grow faster than those of large trees [3,4,14–16]. They argued that lateral growth in large trees may decrease due to increased support costs, hydraulic limitation, or reproductive allocation (associated with branch length, tree height, and stem diameter, respectively). However, we are not aware of any studies that have examined the response of canopy trees by measuring lateral growth both before and after gap formation. Thus, it remains uncertain how gap formation influences the interactions between gap-border trees competing for canopy space.

Canopy disturbance can disrupt the competitive hierarchy established among species of differing life history strategies. It is assumed that gap formation favors species that are less tolerant of shade and regenerate in gaps, because they are better able to exploit the increase in light availability [4,17–19]. For example, yellow birch is able to regenerate in gaps, in part because it has an indeterminate growth pattern that allows it to grow faster than beech and sugar maple, both of which are tolerant of shade and regenerate and survive under closed canopies [20,21]. Thus, it is reasonable to assume that yellow birch trees would respond favorably to gaps because they are better able to exploit the increase in light availability.

Yet, Jones et al. [12] found that yellow birch trees are less responsive to gap formation than either beech or sugar maple. They argued that beech and sugar maple are more responsive because they have low light compensation points, allowing them to be physiologically active for photosynthesis even in shady conditions [22]. Moreover, beech or sugar maple possess deeper crowns and display foliage in the lower part of the crown compared to yellow birch, which often displays foliage in the upper part of the crown [2,12,23]. These crown morphological characteristics allow beech and sugar maple to intercept light along a greater portion of the vertical gap profile, and respond positively to gap formation. Similar positive growth responses of shade-tolerant species have been observed in northern hardwoods [10] and in western conifer stands [9,13]. Thus, gap formation may actually reinforce the competitive hierarchy established among species of contrasting life history strategies, even as it disrupts the asymmetry of competition between large and small trees.

The goal of this study was to examine how gap formation influences the lateral growth of co-occurring tree species with differing autecologies into canopy gaps and to link these growth patterns to competitive interactions among the residual canopy trees. To accomplish this goal, we employed a retrospective approach to quantifying the rate of lateral growth in three hardwood tree species of various shade tolerances (e.g., yellow birch, American beech, and sugar maple), both before and after the formation of small gaps created by single-tree selection. In particular, we attempted to address the following questions regarding the lateral growth of canopy trees bordering the gaps: (1) does gap formation favor intolerant species because they are better able to exploit the increase in light availability? and (2) does gap formation favor small trees because they can grow faster once released from suppression?

2. Materials and Methods

2.1. Study Area

The study was conducted in Haliburton Forest and Wildlife Reserve in Haliburton County, ON, Canada (45°15' N, 78°34' W). The forest is dominated by various tolerant

hardwood species such as sugar maple (*Acer saccharum* Marsh.), representing nearly 60% of the basal area, with a mixture of American beech (*Fagus grandifolia* Ehrh.), yellow birch (*Betula alleghaniensis* Britt.), black cherry (*Prunus serotina* Ehrh.), eastern hemlock (*Tsuga canadensis* (L.) Carr.), and white pine (*Pinus strobus* L.) [24]. The common understory tree species include striped maple (*Acer pennsylvanicum* L.), pin cherry (*Prunus pennsylvanica* L.f.), hobblebush (*Viburnum alnifolium* Michx.), and red raspberry (*Rubus idaeus* L.).

The study area represents a humid continental climate, with a mean annual temperature of 5.0 °C (ranging from −9.9 °C in January to 18.7 °C in July) and seasonal average precipitation of 1074 mm. Soils are classified as Dystric Brunisols that have organic-rich surfaces, and overlie granite–gneiss Precambrian Shield bedrock that is acidic and shallow. The area is characterized by a topography with undulating hills and partially exposed bedrock [25].

Single-tree selection management is the principal mode of silviculture in Haliburton Forest over the last 40 years, creating gaps in the forest canopy of about 0.04 ha in area [26]. Selection cutting is accomplished by conventional logging that removes approximately one third of the basal area every 20 years [27]. This silviculture developed an uneven-aged forest structure, with basal areas ranging from 15 to 30 m² ha^{−1}, while the average canopy height ranged from 20 to 25 m.

2.2. Site and Gap Selection

Information on harvest locations was collected from the Haliburton Forest and Wildlife Reserve Ltd. Based on these records, we chose five stands that were harvested using single-tree selection for sampling in the summer of 2011. During stand selection, efforts were made to locate each stand far from the others. Within each stand, gap locations were chosen by the presence of tree stumps left after harvesting on both sides of the primary skid trails. During gap selection, aspect and direction were not considered, since the variability of light and moisture from the center to the edge of single-tree gaps was assumed to be negligible. However, care was taken so that the gap area had even topography that was accessible by an all-terrain canopy lift (Scanlift 240 manufactured by Kesla, Kesalahti, Finland) used to access the canopy. A total of 31 canopy gaps were sampled.

2.3. Selection of Trees and Branches

In each gap, 3 to 5 trees bordering the gap were chosen that represented three species of differing shade tolerance, namely yellow birch, sugar maple, and American beech. A total of 122 trees were sampled spanning a wide range of diameter classes (DBH ranged from 13 to 67 cm). The majority of the selected trees were sugar maple (79), followed by yellow birch (23) and beech (20). To avoid the effect of long-term canopy openings, trees close to roads or working trails were not selected. The effect of slope position and topography on the selected variables was assumed to be minimal, since the study site was almost flat.

From each selected tree, the longest branch that had grown into the gap was chosen for harvesting by employing a canopy lift. This branch was assumed to represent the tree's maximum crown extent adjacent to the gap, hereafter defined as the sample branch. Efforts were made to select well-lit branches from the upper crown. Each sample branch was harvested with sufficient length for measuring lateral growth before and after the date of harvesting. Before harvesting, the top surface of each branch was marked permanently to correctly position the branch in the laboratory for the purpose of measuring lateral growth. The growth of the sample branch was assumed to represent maximum lateral crown growth into the gap, hereafter referred to as lateral growth or lateral growth rates.

2.4. Tree and Branch Measurements

We measured the DBH of each tree at 1.3 m above the ground using a diameter tape. The total height of each tree (H) was measured using a clinometer. The total horizontal length (m) of each branch (from branch tip to base at tree-bole) was measured using a laser

range finder (distance accuracy ± 10 cm; manufactured by Impulse Laser, Laser Technology Inc., Centennial, CO, USA) from the basket of the lift, hereafter referred to as branch length.

2.5. Measurement of Gap Light Index

To estimate light availability within gaps, hemispherical photographs were taken at the tip of each branch sampled by positioning the lift-basket underneath the branch using a Nikon Coolpix 4500 digital camera equipped with a hemispherical lens (Nikon Fisheye Converter FC-E8 $0.21\times$) attached to a self-leveling gimble with a digital north-finder. All photographs were taken in daylight conditions (a mix of sun and clouds) because of time constraints, albeit the ideal condition for taking photographs is overcast conditions. The photographs were analyzed using a Gap Light Analyzer (GLA) to calculate the light availability in gaps as an index known as gap light index (GLI), which is the percent of direct light transmission through the canopy (GLI ranged between 7% and 98%).

2.6. Data Analysis

To calculate lateral growth rates, we measured the length and angle deflections from horizontal and vertical directions for each internode retrospectively from the date of harvesting following Cole [8]. The horizontal and vertical angles were used to correct the annual internode-extension growth to lateral growth rates. Wherever possible, internode lengths and angle deflections beyond the date of harvesting were measured to calculate lateral growth prior to harvest. Because of the difficulty in correctly identifying internodes, the calculation of preharvest lateral growth rates was mostly restricted to 3 years. However, we were able to correctly calculate preharvest lateral growth rates of up to 10 years.

The pre- and postharvest internode lengths (adjusted by angle deflections) were averaged to obtain the rates of lateral growth before and after harvest. The effect of tree size was examined using stem diameter (DBH) and tree height as predictor variables. We also calculated the change in lateral growth on a per-tree basis by subtracting the average preharvest growth rates from the average postharvest growth rates.

Linear mixed-effect models were employed using R (R Development Core Team, 2011) to determine whether changes in lateral growth and lateral growth rates vary with stem diameter, branch length, tree height, GLI, and species. In addition to fitting the full model, we fit reduced models that included two or three of these variables as predictor(s) of lateral growth rates and changes in lateral growth. Also included in each model was a random unstructured covariance term for stands to account for any unmeasured variances associated with nested data (i.e., trees nested within stands) [28]. To assess the model fit, data of each species were sorted into six diameter bins (depending on the DBH range) to calculate the mean diameter and the mean of the dependent variable predicted for each bin. The fit of these regression models was then compared using Akaike's Information Criterion (AIC) to determine the best model; the model with the lowest AIC score was considered to be the most parsimonious fit [29]. Additionally, the model with a Δ -AIC (difference between the AIC values of two models being compared) of more than 2 was considered better than the model it was compared to. We then used multiple regression analysis to assess the significance of the predictors included in the most parsimonious model. We also used two-sample t-tests to compare the lateral growth before and after gap formation for the significant predictors. Statistical significance was assessed at the 0.05 level ($p < 0.05$).

We conducted a collinearity analysis to evaluate the independence of predictor variables in regressions. Collinearity between predictor variables may prevent them from independently predicting the value of the dependent variable. As such, we assessed collinearity by calculating a tolerance score for each pair of predictor variables; a tolerance score of less than 0.2 was assumed to be indicative of collinearity. Tolerance scores for both height and branch length against stem diameter were below the threshold for multicollinearity (0.2), suggesting that they may have had no effect on lateral growth because they were correlated with stem diameter. However, the tolerance scores for American beech were not presented due to a lack of significance ($p < 0.05$).

3. Results

The average stem diameter of the three hardwood species ranged between 27.9 and 32.2 cm DBH, with considerable variation (SD ranged between 7.1 and 11.4 cm) from tree to tree (Table 1). A similar pattern was observed for tree height (mean and SD varied from 17.6 to 21.3 m and 3.9 to 5.2 m, respectively). Sugar maple had a slightly larger average DBH than American beech and yellow birch (32.1, 27.9, and 28.3 cm, respectively), while American beech was nominally taller on average than yellow birch and sugar maple (21.3, 17.6, and 18.9 m, respectively). However, branch length did not vary among the three species (mean and SD varied from 3.0 to 3.3 m and 0.9 to 1.2 m, respectively).

Table 1. Mean DBH (cm), tree height (m), and branch length (m) for three hardwood tree species bordering single tree gaps.

Species	Mean DBH (SD)	Mean TH (SD)	Mean BL (SD)
Yellow birch	28.33 (11.44)	17.62 (5.22)	3.03 (1.01)
American beech	27.95 (7.11)	21.34 (3.90)	3.43 (0.91)
Sugar maple	32.15 (9.75)	18.94 (4.14)	3.34 (1.29)

TH = tree height; BL = branch length; SD = standard deviation.

There was a large variation in lateral growth among the three species studied (Table 2, Figure 1). Before gap formation, the lateral growth rates were 8.7, 5.2, and 3.3 cm year^{−1} for yellow birch, American beech, and sugar maple, respectively (Figure 1). The lateral growth rates post-gap formation for the three species were 10.1, 7.1, and 4.7 cm year^{−1}, respectively. Thus, yellow birch grew more than twice the rate of sugar maple, both before and after gap formation. Sample branches of American beech grew somewhat faster than sugar maple, but still considerably much slower than yellow birch.

Table 2. Mixed-effects models predicting the change in lateral growth (cm year^{−1}) and lateral growth rates (cm year^{−1}) in three hardwood tree species.

Variable	Model	Parameter					Model Evaluation	
		Species	DBH (cm)	TH (m)	BL (m)	GLI	AIC	Δ-AIC
Change in lateral growth	Model 1 ^a	x *	x *				54.74	0.00
	Model 2	x	x		x		59.63	4.89
	Model 3	x	x			x	65.81	11.07
	Model 4	x	x	x			68.21	13.46
	Full model	x	x	x	x	x	75.92	21.18
Lateral growth	Model 1 ^a	x *	x *				69.93	0.00
	Model 2	x	x		x		82.19	12.26
	Model 3	x	x			x	88.35	18.42
	Model 4	x	x	x			91.77	21.84
	Full model	x	x	x	x	x	96.62	26.69

^a Most parsimonious model fit; x = parameters included in each model; ΔAIC = difference from the most parsimonious model, i.e., model 1; * predictor found significant at $p < 0.05$ in the most parsimonious model through multiple regression analysis; GLI = gap light index.

Yet, the two tolerant species were the most responsive to gap formation (Figure 1). The lateral growth rates of sugar maple and beech increased by 42% and 39%, respectively, both of which were statistically significant ($p < 0.05$). In contrast, yellow birch's growth increased only by 16%, which was not statistically significant.

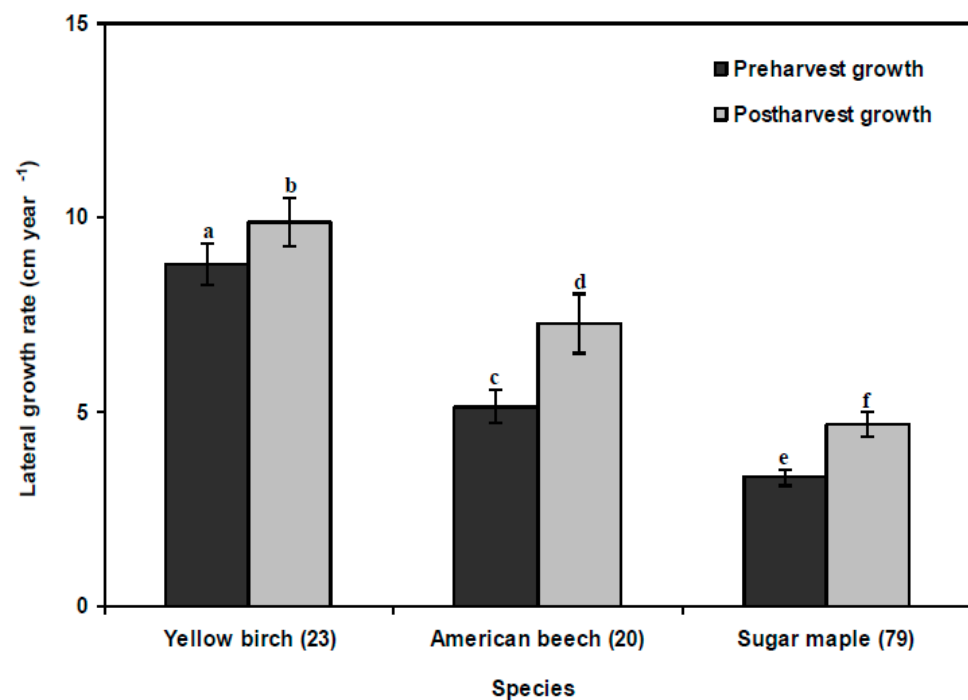


Figure 1. Lateral growth rates (± 2 standard deviations) before and after gap formation in three tolerant hardwood tree species. Numbers in parentheses denote sample size. Bars with different letters in each panel indicate a significant difference at $p < 0.05$.

Lateral growth also varied with stem diameter (Figures 2 and 3; Table 2). Regardless of species, small and large trees grew between 2.8 and 8.1 cm year^{-1} and 3.6 and 9.0 cm year^{-1} , respectively, before gap formation. Thus, small trees grew slower than large trees prior to gap formation (Table 3). However, lateral growth rates after gap formation for the small and large trees varied from 6.4 to 12.9 and 3.7 cm year^{-1} to 8.8 cm year^{-1} , respectively (Table 3). This resulted in the sample branch growth of small trees to increase significantly ($p < 0.05$) by about 5 cm year^{-1} once they were released from suppression (Figure 2). As such, small maple and birch trees grew about twice as fast as their larger counterparts (Figure 3). While small beech trees grew slightly faster than larger ones after gap formation, lateral growth in beech was generally invariant to DBH ($p > 0.05$).

Table 3. Mean lateral growth rates of yellow birch, American beech, and sugar maple, before and after harvest, in three size classes.

Species	Size Class	Lateral Growth Rates (cm Year^{-1})	
		Pre-Harvest $\pm$ Std. Dev.	Post-Harvest $\pm$ Std. Dev.
Yellow birch	Small	(8.1 $\pm$ 0.8) ^a	(12.9 $\pm$ 1.9) ^b
	Medium	(8.9 $\pm$ 0.5) ^a	(8.7 $\pm$ 1.3) ^a
	Large	(9.0 $\pm$ 0.6) ^a	(8.8 $\pm$ 1.4) ^a
American beech	Small	(3.4 $\pm$ 0.4) ^a	(8.0 $\pm$ 0.6) ^b
	Medium	(6.2 $\pm$ 0.2) ^a	(6.6 $\pm$ 0.9) ^a
	Large	(6.1 $\pm$ 0.5) ^a	(6.6 $\pm$ 0.8) ^a
Sugar maple	Small	(2.8 $\pm$ 0.3) ^a	(6.4 $\pm$ 1.0) ^b
	Medium	(3.6 $\pm$ 0.3) ^a	(3.7 $\pm$ 0.5) ^a
	Large	(3.7 $\pm$ 0.2) ^a	(3.9 $\pm$ 0.4) ^a

Note: Size classes include small (10–20 cm DBH), medium (21–30 cm DBH), and large (>30 cm DBH) trees. Means with different letters indicate a significant difference at $p < 0.05$.

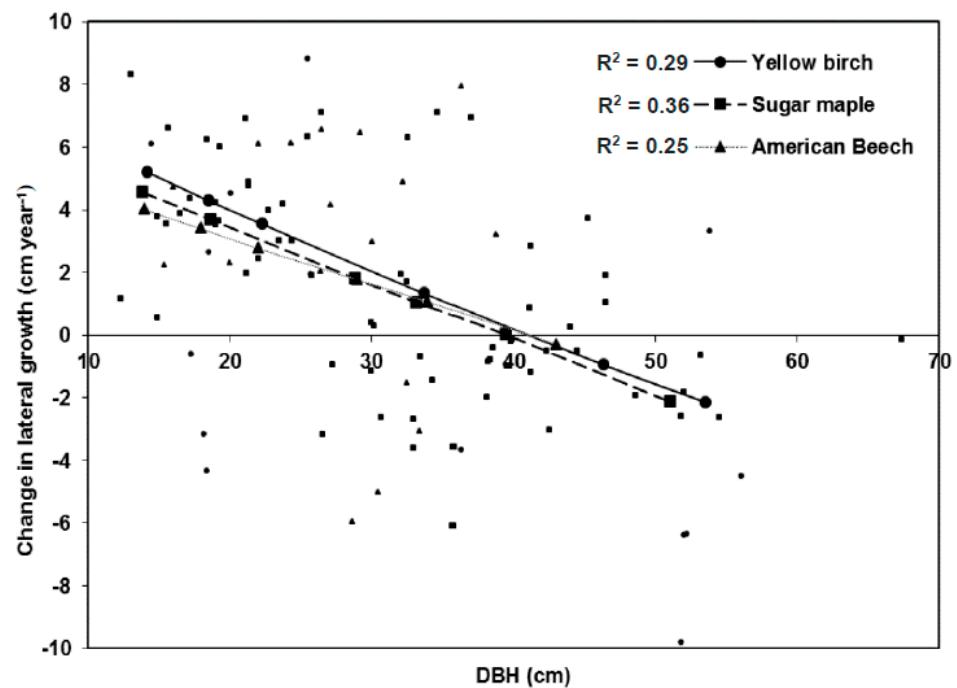


Figure 2. Change in lateral growth in relation to stem diameter (DBH) shown by linear models with coefficient of determination (R^2). To assess the model fit, the data of each species were sorted into six diameter bins (e.g., 10–15, 16–20, 21–25, 26–30, 31–35, and >35 cm) to calculate mean diameter and the mean predicted for each bin.

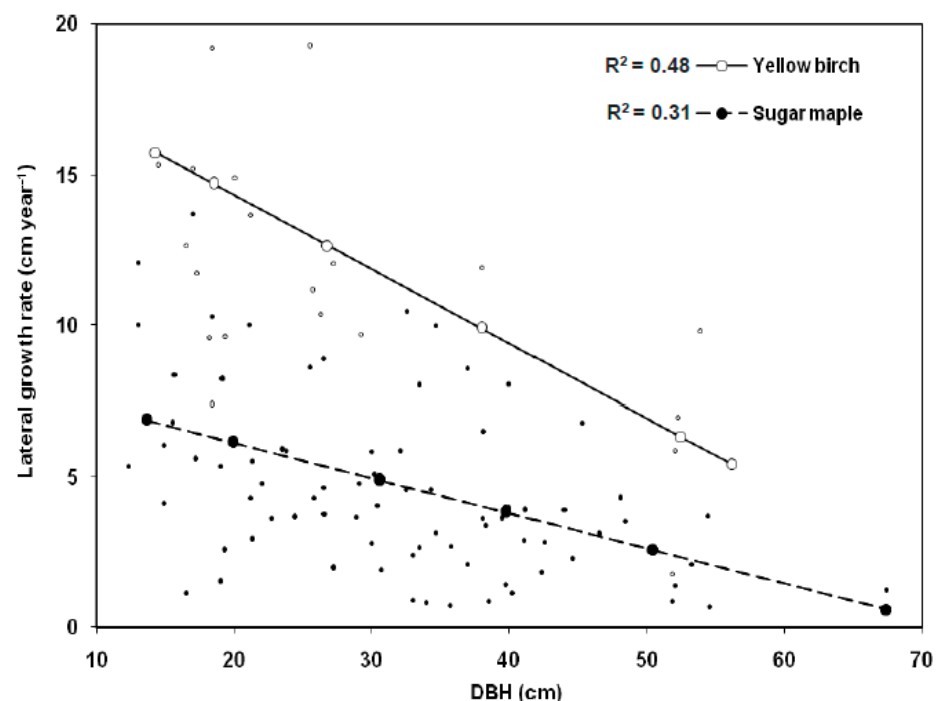


Figure 3. Lateral growth rates after gap formation in relation to stem diameter (DBH) shown by linear models with coefficient of determination (R^2). To assess the model fit, the data of both species were sorted into six diameter bins (e.g., <15, 15–25, 26–35, 36–45, 46–55 and >55 cm) to calculate mean diameter and the mean predicted for each bin. Growth trends for American beech were not shown due to a lack of significance ($p > 0.05$).

The patterns described above were captured by the most parsimonious models, which included both species and DBH as predictors (Table 2). Comparing the AIC values shows that the goodness-of-fit was not improved by adding the other predictors, including height, branch length, and GLI ($\Delta\text{AIC} > 5$ –27). Consequently, multiple regression analyses confirmed that both species and DBH were significant predictors ($p < 0.05$) of growth in the best-fit model, with R^2 values for DBH between species ranging from 0.25 to 0.36 and 0.31 to 0.48, respectively (Figures 2 and 3; Table 2). However, there was no consistent pattern between species showing differences in the strength of variability explained by DBH.

4. Discussion

Gap dynamics can fundamentally shape future forest structure and composition [13,30]. We measured and compared lateral growth before and after the formation of gaps through single-tree selection to understand how canopy disturbance influences the interactions between trees competing for canopy space. In particular, our results pointed to two important aspects of stand dynamics driven by canopy disturbance, namely the size-asymmetry of competition and the co-existence of species.

Numerous studies have suggested that competition for light is highly size-asymmetric, because large trees are able to suppress the growth of smaller trees in the understory by virtue of intercepting a disproportionate share of the incoming light from above [31–34]. The degree of asymmetry has profound consequences for forest dynamics because of its role in accelerating the dynamics of self-thinning and succession, and generating the skewed size distribution characteristic of tree populations [35–37].

However, few studies have examined competition for light between canopy trees [38], so it is uncertain whether height confers any distinct competitive advantage within the canopy itself, particularly mature canopies where smaller trees may take advantage of the gaps left by dead trees. Thus, the ability of smaller trees to grow laterally may be an important means of foraging for light in newly formed gaps, and thereby competing with taller neighbors [15,39–41].

This study demonstrated that gap formation may disrupt the asymmetry of competition between large and small trees, allowing overtopped trees to exploit the increase in light availability by growing laterally into new gaps. For example, the lateral growth of small, overtopped trees increased by 60%–130% (depending on the species), whereas the lateral growth of large trees only increased by 0%–10% (Table 3). Once released from suppression, small trees also generally grew faster than large trees. These results are consistent with prior research that measured stem growth in different forest types [3,4,12–14,42].

There are several different reasons as to why lateral growth may decline as canopy trees mature. First, trees allocate an increasing proportion of resources towards reproduction as they mature [16]. Second, in order to maintain mechanical stability as they grow larger, trees invest an increasing proportion of resources into basal diameter growth, of both the branches and the main stem [43,44]. Third, hydraulic constraints may limit growth as trees grow taller and branches grow longer [45,46].

We found no evidence that lateral growth varied with tree height or branch length, indicating that lateral growth did not decline due to increasing support costs or hydraulic limitation. In contrast, lateral growth did vary with stem diameter, suggesting that growth may decline due to increasing reproductive allocation. Indeed, consistent with this hypothesis, a previous study conducted in the same forest system suggested that ontogenetic increase in reproductive allocation of sugar maple drove the decline in lateral growth [47]. However, these findings may have been the result of collinearity between stem diameter, tree height and branch length. Had there been more orthogonal variation between these predictor variables, we might have observed a significant effect of tree height or branch length on lateral growth.

It is widely accepted that single-tree selection limits the regeneration of species that normally move into the canopy from larger gaps [5,26,48–50]. In the northern hardwood forests of the Great Lakes St. Lawrence region, for example, single-tree selection has likely

contributed to the decline in species such as yellow birch and black cherry that might otherwise coexist with more shade-tolerant species [27,30,51–53]. These species are thought to decline due to a lack of regeneration development in small canopy openings and the accelerated rate of gap closure from lateral expansion of border tree crowns in stands managed using single-tree selection [6,54].

The accelerated closure of gaps in selection-managed stands may occur through two different mechanisms. First, most of the large trees are removed during selection harvest, reducing the average width of tree crowns. As a result, the gaps formed in subsequent harvests become smaller on average [27,49]. Second, by removing the largest trees, selection management favors smaller or younger trees, which are capable of rapid lateral branch extension into newly formed gaps [14]. Together, these changes in stand structure are thought to limit the gap-phase recruitment of mid-tolerant species by reducing the frequency and duration of high-light gaps. The current study confirmed the potential importance of the second mechanism. As discussed above, small trees generally grow faster than large trees, once they gain access to the light in newly formed gaps. For example, sample branches of sugar maple and yellow birch trees less than 15 cm in DBH grow twice as fast as trees greater than 50 cm in DBH. Thus, small silvicultural gaps bordered by small trees may be too ephemeral to permit the regeneration of mid-tolerant species that normally coexist with shade-tolerant species in old-growth forests [6].

This study also supported the notion that selection harvesting may actually reinforce the competitive dominance of shade tolerant species within the canopy [10,13]. The lateral growth of sugar maple and beech increased by 42% and 39%, respectively, whereas the lateral growth of yellow birch only increased by 16%. This was surprising because gap formation generally favors species that regenerate in gaps, because they are better able to exploit the increase in light availability, just as they do in the understory. However, previous research argued that beech and sugar maple are more responsive because they have deeper crowns [2,12,23], allowing them to intercept light along the entire vertical profile of a gap.

In order to promote species co-existence in managed stands, alternative management practices, such as group selection (~4 to 8 trees) or interspersing group- and single-tree cuts that create large openings, are often recommended [6,26,48,55]. Of course, long-term studies examining the relationships between regeneration rates and gap size would be necessary to determine the success of each variant of selection silviculture. For example, group-selection openings with an area from 0.8 ha up to 2.0 ha have been found successful in regenerating mid-tolerant species in northern hardwood forests [55,56]. Alternatively, intermingling single- and group-tree cuts may create gaps large enough to allow saplings of mid-tolerant species to reach the canopy in one gap episode [4]. However, creating larger canopy openings alone may not be enough to overcome all factors that are antagonistic to regenerating diverse tree assemblages, such as intense herbivory and understory competition [57]. Whatever the practice may be, successful establishment of mid-tolerant species requires their immediate response to gaps, which must also be large enough to allow their saplings to move into the canopy in one gap episode. Otherwise, multiple gap episodes must occur through repeat disturbance over a relatively short period of time.

5. Conclusions

In northern hardwood forests managed with single-tree selection, the deliberate creation of small canopy openings can reinforce the competitive dominance of already dominant shade-tolerant species (i.e., sugar maple and American beech). Single-tree selection can also enhance the lateral growth of smaller trees relative to their larger counterparts, thereby potentially disrupting the competitive dominance of large trees in the canopy. Related to this disruption, small silvicultural gaps created by single-tree selection may close more quickly when bordered by small trees than larger trees. Rapid gap closure from lateral branch growth may limit opportunities for regeneration and recruitment of mid-tolerant species in northern hardwood forests, such as yellow birch. If regenerating mid-tolerant

species is a management goal, then selection silviculture may need to be adjusted such that canopy openings become large (often attained via group selection) to increase the probability of recruiting species of lower shade tolerance in northern hardwood forests dominated by very shade-tolerant species.

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